Observe, Imagine, Abstain: Detecting Hallucinated Geometry in Feed-Forward 3D Reconstruction

Anonymous CVPR submission

Abstract

*Feed-forward 3D foundation models reconstruct a scene from unposed photographs in a single pass. Where no photograph shows a surface, they fill the gap from learned priors, and they often assign the invented geometry a confidence indistinguishable from that of observed geometry. We call this geometric hallucination and define it operationally: a predicted point is hallucinated when it lies on a surface that no input view observes. HalluBench-3D produces such labels for free by hiding a 3D region of the ground-truth surface in every input view, so that every pixel inside the region is unsupported and every pixel far from it is supported. We ask whether a model's native confidence separates the two (*AUROC*) and how much unsupported geometry survives a threshold that keeps 90% of supported geometry, the Confident-Unsupported Rate (*CUR*). Across five models (VGGT, Pi3, Pi3X, MapAnything, Depth Anything 3), 18 DTU scans and 18 CO3Dv2 scenes, with five mask placements, two region sizes and three fills each, native confidence lets between 24% and 81% of unsupported pixels through. A realistic inpainted fill makes leakage worse, not better; dropping views or adding a distractor image leaves it unchanged; and the cheapest alternative, variance over perturbed inputs, is worse than native confidence on both backbones we tried. The ranking of models also changes between the lab rig and phone captures: only Pi3 behaves the same on both. We then propose a detector that does not forbid imagination but makes it explicit: a small* support head *on the frozen features of the backbone predicts, per pixel, whether the geometry was observed or imagined, and abstains in between. It costs one backbone forward and is trained on 79 held-out scans with the label the benchmark makes natural: ground-truth multi-view support, the pixels that no other input view sees. On the 18 benchmark scans with 3 to 10 clean views it separates single-view from supported geometry at* AUROC *0.99 against 0.77 to 0.82 for native confidence and 0.86 to 0.88 for a control head that sees one image at a time ($p < 0.0001$ on both backbones); native confidence lets half of the single-view pixels through, the head 2%. On CO3Dv2 phone captures the head never saw, it keeps* AUROC *0.93 on both backbones against 0.70 to 0.76 for native confidence, while the per-image control falls below chance. A head trained on the benchmark's synthetic fills instead scores 1.00, but so does its per-image control, a cautionary result we report in full. Code, benchmark and the full run log will be released.*

1. Introduction

Feed-forward 3D reconstruction has moved from pairwise pointmaps [16, 33] to models that take dozens of unposed images and return dense geometry in one forward pass [15, 19, 31, 34]. These models are trained to output geometry at every pixel, including pixels whose true surface is not visible in any input view. The training objective rewards a plausible guess, so the models learn to guess. The guess is often good, and that is useful. The trouble is that the confidence map, the one signal a user has, does not say which parts were observed and which were guessed.

Suppose you are a conservator looking at a reconstruction of a dome, and you see a crack. Before you send someone up a scaffold, you want to know whether any photograph shows that crack or whether the model drew it from what domes usually look like. Today you cannot tell. The confidence map is bright in both cases.

We make three contributions.

1. **A definition and a benchmark.** We define hallucinated geometry as predicted points on surfaces that no input view shows. HalluBench-3D produces these labels by 3D-consistent masking of ground-truth surfaces, with controls on the same pixels under clean input. We report AUROC, CUR and CUR-ERR for five models on DTU and CO3Dv2, three fill types, an inpainted fill, and stress tests (view dropping, low parallax, a distractor image), each with cluster-bootstrap intervals and paired permutation tests.
2. **A support head that detects missing support.** A head of a few million parameters on the frozen multi-view features of the backbone predicts per pixel whether the output has multi-view support, with an abstain band in between, at no extra backbone forward. Trained on ground-truth visibility of scenes disjoint from the benchmark,



Figure 1. **What a model does with a surface it cannot see.** Top: a DTU scan with one 3D region of the ground-truth surface hidden in all ten input views and inpainted so no seam remains. Bottom: a CO3Dv2 phone capture treated the same way. The top row shows the protocol: the same 3D sphere hides the region in every view, including a view from the other side of the rig. The bottom row shows what the backbone (VGGT) does with it: plausible geometry inside the hidden region, with confidence close to its value outside. The cyan contour marks the unsupported pixels. Model panels are the recorded run outputs.

it passes a pre-registered test against native confidence and against a per-image control (Section 7); trained on the benchmark's synthetic fills instead, it learns the fill, which we report as a warning for masking-based supervision.

3. **Controls that a detector must pass.** Native confidence, perturbation variance, multi-view and reprojection consistency, a linear probe on the same features, and a head that sees only per-image features (a patch detector) are scored on identical rows with paired tests; the support head is also evaluated zero-shot on phone captures and on a fill it never saw.

Contribution 1 is complete and every number in Sections 4 and 5 comes from a logged run. Contributions 2 and 3 are specified in Section 6 with a pre-registered pass condition and evaluated in Section 7; the few cells whose run does not yet exist are marked pending: run id rather than estimated.

The headline claim we test, and do not assume, is that native confidence fails to flag unsupported geometry across current models. It holds, and it holds more strongly on phone captures than on the lab rig.

2. Related work

Feed-forward 3D reconstruction. DUSt3R and MASt3R [16, 33] predict pointmaps for image pairs. VGGT [31], Pi3 and Pi3X [34], MapAnything [15], VGGT-Omega [32] and Depth Anything 3 [19] extend this to many unposed views in one pass. Each exposes a per-pixel confidence trained as a loss weight on observed data; none is trained to distinguish observed from unobserved surfaces. Pi3X replaces Pi3's near-binary confidence with continuous quality levels, so confidence semantics differ even within one family. MapAnything also offers a multi-view consistency confidence, which we include as a baseline.

Hallucination in feed-forward 3D. Bratulic et al. [3] probe DUSt3R, VGGT and DA3 and show that attention heads keep matching correspondences after small patches are occluded, concluding that these models hallucinate missing correspondences from learned priors; they measure attention and Sampson error on ShapeNet pairs, not confidence or 3D labels. Paul et al. [25] benchmark hallucination at the input level: unrelated scenes, repeated images and noise still produce dense geometry, scored by COLMAP-based consistency. 3D-Mirage [23] studies monocular depth under visual illusions. Li et al. [17] show that wide viewpoint spans produce unsupported geometry, and Dadajonov and Umarhojiyev [6] that leaving parts of a room unobserved is not revealed by inference-time uncertainty; neither has per-point labels. FreeInpaint [24] fills masked regions on purpose, the mechanism these models apply unasked. Our definition is per point, with exact supported and unsupported labels, and it tests whether native confidence separates them.

Uncertainty for feed-forward 3D. ReVar3R [20] takes per-point variance over perturbed runs; it beats native confidence on AUSE in most conditions but, by its authors' account, misses stable systematic bias, which is what a confidently filled region is. Trust3R [35] fits an evidential head on a frozen backbone, GeoCond [1] adds pose-level uncertainty, and Deng et al. [9] find the implied uncertainty of seven backbones at least $2.4\times$ too small. All score uncertainty against ground-truth error, which conflates confident errors on observed surfaces with confident predictions on surfaces no view saw; CUR isolates the second case. Hillemann et al. [10] score VGGT's native uncertainty against reconstruction error on DTU. Closest to our detector, a linear probe on frozen VGGT features predicts *camera-pose* failure with AUROC 0.935 against 0.797 for native confidence [30], and a small head on the same features predicts pairwise co-visibility [36]; our support head is the dense,

per-pixel analogue for unobserved *geometry*, trained without labels, and the linear probe is one of our baselines. Free Geometry [7] and TTT3R [4] adapt at test time, but neither labels unsupported regions.

Generative and multi-sample 3D. Z3D [2] decodes hidden surfaces from frozen VGGT features, evidence that unobserved geometry is present in them. Feed-forward Gaussian splatting from unposed views [12] outputs pixel-aligned Gaussians with opacities, and ReconSplat [28] extrapolates beyond the observed views by design; a model built to extrapolate has all the more reason to say where it did. Real-time world models now keep a persistent 3D Gaussian memory updated from the frames they generate [5]; such a memory stores imagined and observed surfaces alike, which is exactly where a per-point support signal belongs. VolFill [22] finds sample variation $1.3\times$ larger in occluded regions; diffusion depth ensembles [13, 14] give sample-disagreement uncertainty; a generative head over quantized geometry [8, 21] whose samples disagree where the model imagines is the natural successor to our discriminative head, and future work.

Hallucination in vision-language models. POPE [18] labels statements unsupported by the image; ours is the geometric analogue, geometry unsupported by any view.

3. Defining geometric hallucination

Given input views $I_1, \ldots, I_N$, a model returns for each view i a depth map D_i and a confidence map C_i. A pixel p in view i is *supported* if its ground-truth surface point is visible in at least one input view, and *unsupported* if it is visible in none. A model hallucinates at p when it outputs geometry at an unsupported p with high confidence.

Why confidence is the wrong tool. Native confidence is trained as a per-pixel loss weight on observed data. It learns to be low where the regression target is noisy (edges, specular surfaces) and high where it is smooth. A region that no view sees and that the prior fills smoothly therefore receives high confidence by construction. Section 5 measures how large this effect is.

Metrics. All are computed per (model, scene, seed, radius, fill) and averaged with cluster-bootstrap intervals over scenes.

- AUROC: $P(C_S > C_U)$ with ties counted half. 1.0 means every supported pixel outranks every unsupported one; 0.5 means confidence carries no information about support.
- CUR: choose τ so that 90% of supported pixels have $C \geq \tau$; CUR is the fraction of unsupported pixels with $C \geq \tau$. This is the hallucination rate you would experience after filtering by confidence.
- CUR-ERR: the fraction of unsupported pixels above τ whose AbsRel exceeds 0.1. Confident and wrong.
- AbsRel_U, AbsRel_S: depth error on unsupported and supported pixels after median scaling fitted on supported pixels only.
- Controls: the same pixels under clean input (AbsRel_U ctrl, AUROC ctrl) and the ratio of median confidence inside the hidden region to its clean value.

4. HalluBench-3D

Protocol. For each scene we take 10 views. A sphere is centred on a random ground-truth surface point in the reference view, with radius 0.10 or 0.18 of the object's bounding-box diagonal, resampled until it covers 2 to 35% of the reference view. Every pixel in every view whose ground-truth point lies inside the sphere is hidden (Figure 2). Hidden pixels are filled with gray, a blur of the original, uniform noise, or an inpainting of the surround [29]; fill type is an ablation so that results are not an artefact of a visible patch. Labels: U is hidden in all views; S is valid and outside a 6-pixel dilation of the hidden region; the ring in between is ignored. Five mask seeds per scene; masks are deterministic in (scene, seed, radius).

Datasets. *DTU* [11]: the 18 scans of the MVSNet validation split, 10 views spread over the 49-camera rig, model input 420×518. *CO3Dv2* [26]: 18 scenes, the two depth-carrying sequences of each of nine categories (apple, ball, bench, hydrant, mouse, teddybear, toytrain, toytruck, vase) from the single-sequence subset, with the official COLMAP depth as ground truth. The 10 views are spread over the full orbit, so the hidden region is seen from the back as well as the front. Frames are 640×360 phone video; models run at 518×294 (or the transpose for landscape sequences). Everything else, including every threshold, is unchanged from DTU. *ETH3D* [27]: nine high-resolution DSLR training scenes (courtyard, facade, relief, terraces, terrains) at metre to tens-of-metre scale with laser-scanner ground truth, resampled onto the undistorted pinhole frames; frames 640×427, models 518×350; the reference frame and its nine most co-visible frames. These are the building-scale scenes the first two datasets lack. Training-data overlap: VGGT, Pi3/Pi3X and Depth Anything 3 list CO3Dv2 among their training sets [19, 31, 34]; MapAnything does not [15]; none of the four lists DTU or ETH3D. CO3Dv2 is therefore in-distribution for three of the backbones, and DTU and ETH3D are out of distribution for all of them.

Correctness guards. Before any model is scored we check that ground-truth depth reprojects consistently be-

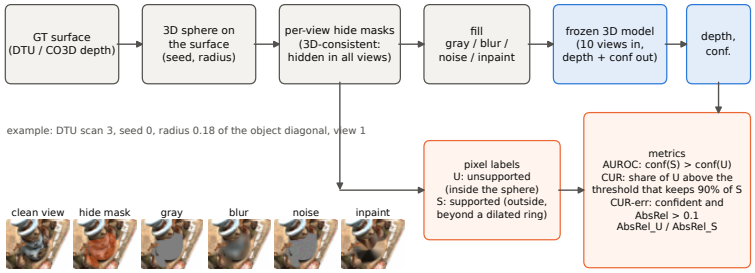


Figure 2. **The benchmark protocol.** A sphere is placed on the ground-truth surface; every pixel in every view whose ground-truth point falls inside it is hidden, so the hidden region is 3D-consistent. Hidden pixels are filled (gray, blur, noise or inpaint), the model runs on the filled views, and its depth and confidence are scored against the unsupported (U), supported (S) and ring labels. The same pixels are also scored under clean input as a control.

tween view pairs and that unsupported points of view i, where visible in view j, land inside view j's hidden mask. DTU: reprojection agreement 0.80 to 0.82 at a 1% depth tolerance, U-transfer 0.96 to 0.99. CO3Dv2 (cameras converted from PyTorch3D and checked against the official point cloud): with an occlusion-aware 3% tolerance, agreement 0.92 to 1.00 and U-transfer 0.90 to 0.98 on 17 of 18 scenes; the eighteenth scores 0.80, is kept, and removing it changes nothing (Section 5). ETH3D: agreement 0.996 to 1.000 on all scenes; nine of twelve pass the U-transfer gate and three are excluded before any model is run.

Pipeline validation. On a synthetic DTU-format scene, an oracle that knows what is hidden scores AUROC 1.000 and CUR 0.000, and a blind guesser with flat confidence scores AUROC 0.500 and CUR 0.90 (the 10% of supported pixels dropped by τ is the expected floor). These numbers validate the code, not any model.

Models. VGGT-1B [31] (depth and depth confidence), Pi3 and Pi3X [34] (local point z and sigmoid confidence), MapAnything [15] (image-only metric mode), Depth Anything 3 [19] (Large 1.1), and AnySplat [12], a feed-forward Gaussian-splatting model scored on each pixel's Gaussian depth with its opacity as confidence (pending: P3-gs), to ask whether the representation changes the answer. Same views and resolution for all; fp16 on a T4 (AnySplat bf16 on an L4). VGGT-Omega is gated and not yet run.

Statistics. Means carry 95% cluster-bootstrap intervals over scenes (10,000 resamples; each scene contributes its mean over seeds). Pairwise comparisons use a paired sign-flip permutation test on per-scene means (20,000 permutations), Holm-corrected within each family of pairs.

5. Results on HalluBench-3D

5.1. DTU: native confidence leaks

Table 1 gives the full DTU configuration (the per-run spread is in the supplementary material). Pooled over the two natural-statistics fills (blur and noise), CUR is 0.242 for VGGT, 0.281 for Pi3, 0.532 for Pi3X, 0.622 for MapAnything and 0.701 for Depth Anything 3, with AUROC 0.903, 0.912, 0.813, 0.743 and 0.711. Three tiers emerge. VGGT and Pi3 drop their confidence inside the hidden region (to 19 to 41% of its clean value) yet still let one in four unsupported pixels through a threshold that keeps 90% of supported ones. Pi3X leaks one in two, and under a noise fill it is as confident about geometry it cannot see as about geometry it can (ratio 1.09). MapAnything and Depth Anything 3 are close to blind to non-observation. DA3's confidence inside the hidden region stays at 80 to 86% of its clean value for every fill, including the gray patch; whatever its confidence encodes, it is not observation.

Pairwise, after Holm correction over the ten pairs: VGGT and Pi3 are indistinguishable ($p = 0.41$); both leak less than Pi3X ($p \leq 0.0015$), MapAnything and DA3 ($p = 0.0005$). Pi3X versus DA3 ($p = 0.052$), Pi3X versus MapAnything and MapAnything versus DA3 ($p = 0.19$) are not separated.

Confident and wrong. On DTU the imagined depth is accurate for four of the five models (ABSREL$_U$ 0.011 to 0.022, about 1.3 to 2× the clean value), so CUR-ERR stays near zero. The failure is that you cannot tell imagined from observed, not that the imagination is poor. MapAnything is the exception: 29% of its unsupported pixels are confident and wrong by more than 10% ABSREL, on top of a 10% error on supported pixels in image-only metric mode (unchanged at its native input size).

Table 1. **DTU, five models, full configuration.** 18 scans × 5 mask seeds × 2 radii × 3 fills, pooled over radii; mean [95% cluster-bootstrap CI over scans]. *conf ratio* is the median confidence inside the hidden region divided by its value under clean input. Runs P1-full and P2-full, T4, fp16.

model	fill	AUROC	CUR	CUR-ERR	AbsRel$_U$	AbsRel$_U$ ctrl	conf ratio
VGGT-1B	gray	0.973 [0.955, 0.987]	0.067 [0.031, 0.111]	0.000	0.018	0.006	0.051
VGGT-1B	blur	0.886 [0.843, 0.922]	0.263 [0.203, 0.324]	0.000	0.011	0.006	0.241
VGGT-1B	noise	0.919 [0.892, 0.945]	0.222 [0.145, 0.304]	0.000	0.014	0.006	0.192
Pi3	gray	0.981 [0.972, 0.988]	0.053 [0.025, 0.085]	0.001	0.020	0.014	0.057
Pi3	blur	0.923 [0.895, 0.947]	0.236 [0.156, 0.326]	0.001	0.018	0.014	0.261
Pi3	noise	0.902 [0.865, 0.934]	0.325 [0.219, 0.438]	0.002	0.020	0.014	0.411
Pi3X	gray	0.940 [0.902, 0.971]	0.202 [0.085, 0.337]	0.002	0.022	0.017	0.335
Pi3X	blur	0.826 [0.782, 0.865]	0.512 [0.397, 0.630]	0.002	0.020	0.017	0.739
Pi3X	noise	0.801 [0.734, 0.864]	0.551 [0.390, 0.711]	0.004	0.022	0.017	1.091
MapAnything	gray	0.861 [0.812, 0.905]	0.374 [0.272, 0.480]	0.195 [0.125, 0.269]	0.222	0.111	0.650
MapAnything	blur	0.773 [0.720, 0.825]	0.580 [0.482, 0.672]	0.277 [0.197, 0.361]	0.135	0.111	0.677
MapAnything	noise	0.712 [0.645, 0.773]	0.664 [0.566, 0.758]	0.295 [0.211, 0.390]	0.129	0.111	0.709
Depth Anything 3	gray	0.723 [0.664, 0.780]	0.703 [0.596, 0.804]	0.003	0.017	0.013	0.804
Depth Anything 3	blur	0.682 [0.611, 0.744]	0.711 [0.619, 0.796]	0.003	0.016	0.013	0.800
Depth Anything 3	noise	0.740 [0.686, 0.793]	0.690 [0.578, 0.793]	0.001	0.017	0.013	0.855

Gate. Before scaling the benchmark we fixed a pass condition: CUR above zero with intervals excluding zero, and AUROC clearly below one on the natural fills. It passes for all five models. The effect is not an artefact of a gray patch, and it is strongly model dependent.

5.2. Stress tests: the leak is not a patch artefact

A reasonable objection to Table 1 is that a filled patch is visible, and the model is reacting to the patch rather than to the absence of a surface. Table 2 and Figure 3 answer it four ways.

Fill realism. An inpainted fill is edge-continuous and has no texture discontinuity. It raises leakage further for VGGT (+0.088, $p = 0.031$) and Pi3X (+0.068, $p = 0.005$). Table 1 is therefore a lower bound on what a natural-looking occluder produces.

Evidence quantity. Dropping from 10 to 3 views leaves CUR unchanged for all three models. Having fewer views to cross-check does not make the models less sure, which says that native confidence is driven by single-view appearance rather than multi-view agreement.

Geometry under stress. With 3 views or low parallax, Pi3X's imagined depth degrades (AbsRel$_U$ 0.05, CUR-ERR up to 0.04) while its confidence does not move. The confident-and-wrong fraction, invisible on the easy rig, appears under harder capture.

DA3 is different. A distractor image raises its leakage and low-parallax input halves it, consistent with a confidence that tracks appearance consistency across views rather than observation of the surface. We flag this as an observation to explain, not as an explanation.

5.3. CO3Dv2: the ranking changes

Table 3 and Figure 4 repeat the full protocol on CO3Dv2 phone captures, with every threshold frozen from DTU. Pooled over blur and noise, CUR is 0.272 for Pi3, 0.570 for Pi3X, 0.651 for VGGT, 0.797 for DA3 and 0.810 for MapAnything; AUROC is 0.891, 0.732, 0.681, 0.620 and 0.576.

The ranking changes. Pi3 is the only model whose behaviour transfers from the lab rig to phone video (CUR 0.27 on both). The leak is larger on the dataset three of the models were trained on: seeing CO3Dv2 at training time did not teach their confidence to notice a missing surface. VGGT, joint best on DTU, leaks 65% of unsupported pixels on CO3Dv2, 2.7× its DTU value, and under a noise fill its confidence is almost blind to non-observation (AUROC 0.59). MapAnything and DA3 let about four in five unsupported pixels through. After Holm correction, Pi3 leaks less than every other model ($p = 0.0005$ each); Pi3X leaks less than MapAnything and DA3 ($p = 0.0005$); VGGT leaks less than MapAnything ($p = 0.010$) and DA3 ($p = 0.0005$); VGGT versus Pi3X ($p = 0.23$) and MapAnything versus DA3 ($p = 0.73$) are not separated. Across datasets, the intervals of VGGT, MapAnything and DA3 do not overlap; those of Pi3 and Pi3X do.

Imagination is accurate here too. AbsRel$_U$ is within 0.005 of AbsRel$_S$ for every model, so CUR-ERR stays below 0.04, and MapAnything's confident-and-wrong behaviour from DTU does not appear (AbsRel$_S$ 0.041, CUR-ERR 0.007). These are everyday objects with strong shape priors; the models complete them well and report the completion with the confidence of an observation.

Fill effects differ from DTU. On DTU every fill raised CUR over gray for VGGT, Pi3 and Pi3X. On CO3Dv2 noise raises VGGT's CUR by 0.20 ($p = 0.00005$) but blur does not ($p = 0.30$); Pi3X reacts to both (+0.26, +0.23); Pi3 weakly (+0.06, +0.10); MapAnything and DA3 not at all (p from 0.06 to 0.91). A confidence that does not change when the patch changes is not reading the patch.

Table 2. **Stress tests on DTU.** 18 scans × 2 mask seeds, radius 0.18, models VGGT, Pi3X, DA3. Reference rows are the matching cells of Table 1 (blur, 10 spread views, seeds 0 and 1). Last column: paired CUR difference against the reference with a sign-flip p over scans. Run P2-stress.

model	test	AUROC	CUR	CUR-ERR	ABSREL_U	CUR vs ref
VGGT-1B	reference	0.902 [0.868, 0.930]	0.259 [0.175, 0.349]	0.000	0.010	
VGGT-1B	inpaint fill	0.867 [0.821, 0.907]	0.347 [0.255, 0.443]	0.001	0.018	$+0.088$ ($p = 0.031$)
VGGT-1B	5 views	0.902	0.269 [0.173, 0.377]	0.000	0.011	$+0.010$ ($p = 0.63$)
VGGT-1B	3 views	0.899	0.259 [0.167, 0.361]	0.000	0.013	$+0.001$ ($p = 0.98$)
VGGT-1B	low parallax	0.911	0.214 [0.110, 0.334]	0.000	0.021	-0.045 ($p = 0.35$)
VGGT-1B	+ distractor	0.905	0.257 [0.179, 0.340]	0.000	0.011	-0.002 ($p = 0.90$)
Pi3X	reference	0.828 [0.781, 0.870]	0.521 [0.390, 0.650]	0.002	0.024	
Pi3X	inpaint fill	0.786 [0.731, 0.836]	0.589 [0.473, 0.702]	0.004	0.022	$+0.068$ ($p = 0.005$)
Pi3X	5 views	0.822	0.539 [0.411, 0.661]	0.006	0.024	$+0.018$ ($p = 0.44$)
Pi3X	3 views	0.840	0.497 [0.373, 0.618]	0.032	0.051	-0.024 ($p = 0.64$)
Pi3X	low parallax	0.812	0.523 [0.376, 0.671]	0.040	0.050	$+0.002$ ($p = 0.98$)
Pi3X	+ distractor	0.829	0.510 [0.368, 0.650]	0.002	0.024	-0.011 ($p = 0.66$)
DA3	reference	0.712 [0.632, 0.783]	0.679 [0.569, 0.791]	0.006	0.018	
DA3	inpaint fill	0.726	0.688 [0.605, 0.774]	0.009	0.018	$+0.008$ ($p = 0.87$)
DA3	5 views	0.697	0.785 [0.703, 0.865]	0.005	0.019	$+0.106$ ($p = 0.008$)
DA3	3 views	0.729	0.667 [0.560, 0.770]	0.018	0.031	-0.012 ($p = 0.81$)
DA3	low parallax	0.863 [0.785, 0.925]	0.372 [0.230, 0.520]	0.008	0.029	-0.307 ($p = 0.0003$)
DA3	+ distractor	0.660 [0.580, 0.734]	0.775 [0.680, 0.868]	0.003	0.019	$+0.096$ ($p = 0.0004$)

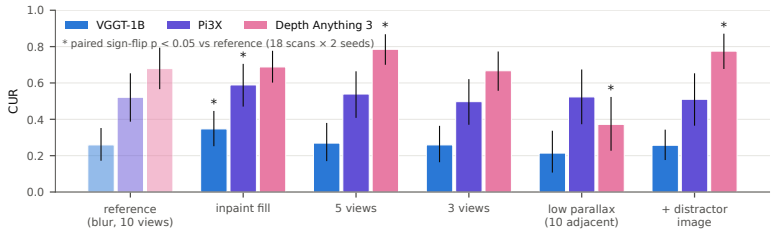


Figure 3. **Stress tests.** CUR under each stress condition against the reference, with intervals. View count does nothing for VGGT and Pi3X; the inpainted fill raises leakage; DA3 alone reacts to parallax and to a distractor.

Per-scene view. Figure S3 in the supplementary shows that the leak is not driven by a few scenes. Removing the one scene that missed the mask-transfer guard moves every pooled number by at most 0.013.

5.4. Cheap detectors do not help

Two obvious alternatives are in Table 4. Variance over $K = 4$ photometrically perturbed runs, aligned per sample by median scale, is much worse than native confidence on both backbones (paired $p \leq 0.0007$ in all four tests) while costing four forwards: the imagined geometry is stable under input perturbation, the systematic bias ReVar3R's authors describe. Reprojection consistency with the model's own cameras is no better than chance (AUROC 0.49, 0.48): the imagined surface is cross-view coherent. MapAnything's multi-view consistency confidence leaks more than its learned confidence (CUR 0.51 against 0.33 in the smoke configuration).

6. Method: observe, imagine, abstain

The benchmark says that native confidence does not know what the model has seen. Rather than suppress the prior, which is what makes these models useful, we want the model to say which of its outputs came from the prior. Figure 5 shows the design.

Architecture. A frozen backbone (VGGT or Pi3X) provides geometry, native confidence, and the multi-view features that produced them: for VGGT the aggregated tokens of the four layers its depth head reads (2048-d at 14×14 patches), for Pi3X the 2048-d decoder tokens feeding its confidence head. The *support head* (LayerNorm and 1×1 projection per layer to 256 channels, sum, two 3×3 convolutions at patch resolution, one logit per patch upsampled bilinearly) outputs $P(\text{imagined})$; the benchmark scores $1 - P(\text{imagined})$. A few million parameters; the backbone is neither modified nor re-run.

Training labels. Two label sources, both on the 79 scans of the DTU training split, disjoint from the 18 benchmark scans and the 22 extension scans of Section 7. *Visibility labels* (the main head): each pixel's ground-truth point is projected into the other input views; a pixel that no other view sees, with no view undecided, is *single-view*, a pixel seen by two or more is *supported* (3% depth tolerance, a 3-pixel margin). The views are clean, 3, 5 or 10 at ran-

Table 3. **CO3Dv2, five models, full configuration.** 18 scenes × 5 seeds × 2 radii × 3 fills, pooled over radii; mean [95% CI over scenes]. Runs P2-co3d and P2-co3d-b.

model	fill	AUROC	CUR	CUR-ERR	AbsRel$_S$	AbsRel$_U$	AbsRel$_U$ ctrl	conf ratio
VGGT-1B	gray	0.747 [0.692, 0.802]	0.565 [0.453, 0.677]	0.029	0.029	0.031	0.027	0.634
VGGT-1B	blur	0.772 [0.717, 0.825]	0.534 [0.416, 0.652]	0.034	0.029	0.031	0.027	0.603
VGGT-1B	noise	0.590 [0.517, 0.656]	0.767 [0.693, 0.839]	0.029	0.029	0.029	0.027	0.805
Pi3	gray	0.930 [0.894, 0.958]	0.196 [0.118, 0.283]	0.002	0.015	0.016	0.013	0.351
Pi3	blur	0.903 [0.860, 0.939]	0.251 [0.170, 0.334]	0.002	0.015	0.016	0.013	0.389
Pi3	noise	0.880 [0.813, 0.929]	0.293 [0.193, 0.405]	0.002	0.015	0.015	0.013	0.472
Pi3X	gray	0.863 [0.825, 0.899]	0.329 [0.248, 0.408]	0.001	0.017	0.022	0.014	0.749
Pi3X	blur	0.720 [0.661, 0.777]	0.584 [0.500, 0.668]	0.002	0.017	0.018	0.014	0.877
Pi3X	noise	0.743 [0.680, 0.798]	0.556 [0.462, 0.650]	0.001	0.017	0.019	0.014	0.867
MapAnything	gray	0.609 [0.554, 0.665]	0.786 [0.721, 0.849]	0.004	0.041	0.029	0.034	0.632
MapAnything	blur	0.575 [0.514, 0.637]	0.801 [0.735, 0.863]	0.008	0.042	0.028	0.034	0.674
MapAnything	noise	0.578 [0.524, 0.631]	0.819 [0.765, 0.870]	0.007	0.041	0.033	0.034	0.690
Depth Anything 3	gray	0.619 [0.560, 0.674]	0.816 [0.751, 0.877]	0.001	0.019	0.015	0.014	0.874
Depth Anything 3	blur	0.622 [0.574, 0.668]	0.813 [0.748, 0.872]	0.001	0.018	0.015	0.014	0.853
Depth Anything 3	noise	0.618 [0.553, 0.679]	0.780 [0.702, 0.854]	0.000	0.019	0.016	0.014	0.901

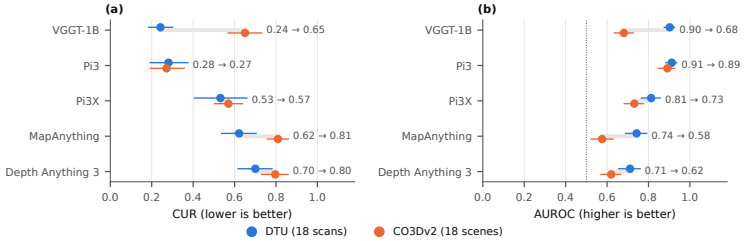


Figure 4. **Lab rig versus phone captures.** Pooled blur-and-noise CUR and AUROC per model on DTU and CO3Dv2 with 95% intervals. Pi3 is the only model whose numbers transfer; VGGT, MapAnything and DA3 leak markedly more on CO3Dv2.

dom per step; the head learns to fire on single-view pixels, the benchmark's notion of support without the patch, at the price of needing multi-view ground truth. *Self-labels* (the fill-trained variant): the backbone is run once on clean views, a sphere is placed on its *predicted* surface and projected with its *predicted* depth and cameras into every view, the views are filled (gray, blur or noise; inpaint held out) and the head learns to find the hidden pixels, with 15% clean samples. Both: class-balanced binary cross-entropy, 1500 AdamW steps (learning rate 10^{-3}, cosine, warm-up 100), one scene of random views per step, 1.2 to 2.5 hours per backbone on one L4 or T4; hyperparameters fixed before any benchmark scan was evaluated.

Inference and compute. One backbone forward, one head forward; the head adds under 1% of the backbone's FLOPs, against four backbone passes for perturbation variance. Three colours for the user: *observed* above $1 - P(\text{imagined}) = 0.7$, *imagined* below 0.3, *abstain* between (thresholds fixed in advance); the metrics use the continuous score.

Controls and ablations. A head could learn an image cue (the patch; image borders and depth edges, where single-view pixels concentrate) rather than the missing support. The decisive control is the *per-image feature head*: the same head on per-image encoder tokens before any cross-view attention, trained on the same labels. Further controls: the held-out inpainted fill and CO3Dv2 for the fill-trained head, a *linear probe* on the same features [30], extra seeds, a transformer head and fusion with native confidence; heads sharing a label source train jointly on the same backbone forwards.

7. Experiments on the method

Fill-trained head: what it learns. Its pre-registered test: (1) beat native confidence on AUROC and CUR (paired sign-flip $p < 0.05$, 18 scans, one forward); (2) the per-image control does not explain the gain on the inpainted fill or on CO3Dv2. Part 1 passes for both backbones ($p = 0.00005$; AUROC 0.990 to 1.000 on every cell of the full configuration and the stress tests). Part 2 fails in substance: the control matches the head on every fill, on the held-out inpainted fill and on CO3Dv2 (VGGT 0.992 [0.987, 0.996] against 0.989 [0.985, 0.993]; Pi3X head 0.975,

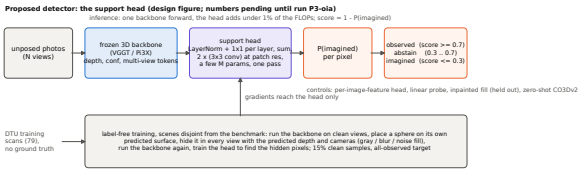


Figure 5. **Proposed detector: the support head.** A frozen 3D backbone supplies geometry, native confidence and the multi-view tokens that produced them; a small head on those tokens outputs $P(\text{imagined})$ per pixel, shown to the user as *observed*, *abstain* or *imagined*. Shown is the label-free variant: the backbone is run on clean views, a sphere is placed on its *own* predicted surface and hidden in every view, and the head learns to find the hidden pixels. The main head is trained instead on ground-truth visibility, the pixels no other input view sees, from clean views and without a patch. Training scenes are disjoint from the benchmark. This figure specifies the design; Section 7 reports what the head learned.

Table 4. **Detectors on DTU** (18 scans, seed 0, blur, radius 0.18; native and perturbation rows re-run on the detectors' hardware, runs P3-native and P3-pv, so every pair is scored on identical rows). Every cell comes from a logged run.

backbone	detector	fwd	AUROC	CUR
VGGT-1B	native confidence	1	0.915 [0.883, 0.943]	0.255 [0.157, 0.364]
VGGT-1B	perturbation var., K=4	4	0.666 [0.602, 0.729]	0.627 [0.521, 0.732]
VGGT-1B	reprojection consistency	1	0.488 [0.473, 0.506]	0.931 [0.903, 0.954]
VGGT-1B	linear probe on features	1	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
VGGT-1B	per-image feature head (control)	1	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
VGGT-1B	**support head**	1	1.000 [1.000, 1.000]	0.000 [0.000, 0.000]
Pi3X	native confidence	1	0.832 [0.782, 0.879]	0.502 [0.362, 0.642]
Pi3X	perturbation var., K=4	4	0.550 [0.491, 0.608]	0.824 [0.754, 0.886]
Pi3X	reprojection consistency	1	0.478 [0.435, 0.518]	0.929 [0.899, 0.955]
Pi3X	per-image feature head (control)	1	1.000 [0.999, 1.000]	0.000 [0.000, 0.001]
Pi3X	**support head**	1	0.999 [0.999, 1.000]	0.001 [0.000, 0.003]

Table 5. **True-visibility protocol on DTU and CO3Dv2.** DTU: 18 scans, clean inputs of 3, 5 and 10 views, three view sets each, pooled; single-view pixels are 18%, 9% and 2.5% of the known pixels. CO3Dv2: 18 phone-video scenes the head never saw, 3 and 5 views pooled (10 views around an object leave almost no single-view pixel, and the protocol skips the scene); single-view pixels are 10% and 2.5%. AUROC and CUR for separating single-view from supported pixels; AbsRel of the prediction on single-view pixels is about twice that on supported ones (DTU: VGGT 0.014 against 0.006, Pi3X 0.036 against 0.018; CO3Dv2: 0.036 against 0.024, 0.028 against 0.015). Runs P4-vis and P4-vis-co3d.

backbone	detector	AUROC	CUR
VGGT-1B	native confidence	0.772 [0.722, 0.823]	0.530 [0.443, 0.616]
VGGT-1B	reprojection consistency	0.954 [0.944, 0.963]	0.082 [0.067, 0.098]
VGGT-1B	per-image feature head (control)	0.878 [0.845, 0.907]	0.340 [0.289, 0.394]
VGGT-1B	linear probe on features	0.989 [0.985, 0.992]	0.024 [0.015, 0.035]
VGGT-1B	**support head**	0.993 [0.990, 0.995]	0.015 [0.008, 0.024]
Pi3X	native confidence	0.820 [0.793, 0.845]	0.493 [0.435, 0.553]
Pi3X	reprojection consistency	0.914 [0.900, 0.929]	0.164 [0.120, 0.213]
Pi3X	per-image feature head (control)	0.863 [0.831, 0.891]	0.375 [0.326, 0.425]
Pi3X	linear probe on features	0.985 [0.979, 0.989]	0.034 [0.021, 0.049]
Pi3X	**support head**	0.990 [0.986, 0.994]	0.021 [0.010, 0.035]
CO3Dv2, never seen by the head (3 and 5 views)			
VGGT-1B	native confidence	0.755 [0.715, 0.793]	0.722 [0.649, 0.795]
VGGT-1B	reprojection consistency	0.794 [0.747, 0.841]	0.688 [0.505, 0.858]
VGGT-1B	per-image feature head (control)	0.461 [0.389, 0.540]	0.879 [0.813, 0.933]
VGGT-1B	linear probe on features	0.915 [0.885, 0.943]	0.248 [0.164, 0.335]
VGGT-1B	**support head**	0.929 [0.905, 0.952]	0.208 [0.138, 0.284]
Pi3X	native confidence	0.697 [0.649, 0.740]	0.846 [0.802, 0.888]
Pi3X	reprojection consistency	0.888 [0.863, 0.910]	0.219 [0.160, 0.287]
Pi3X	per-image feature head (control)	0.426 [0.356, 0.507]	0.889 [0.823, 0.941]
Pi3X	linear probe on features	0.933 [0.910, 0.955]	0.200 [0.131, 0.273]
Pi3X	**support head**	0.932 [0.909, 0.954]	0.219 [0.135, 0.309]

control pending: P3-oia-co3d). The clause "head ahead at $p < 0.05$" is formally met by differences of 0.0004 and 0.003 while the control carries 99% of the gain: a flaw of that criterion, not a pass. A head trained on the benchmark's fills learns the fill, detectable from one image even when inpainted. The per-image control is mandatory for anyone training on masking labels.

Visibility-trained head: the real test. Gate 3 was fixed before this head was trained, with a margin this time: beat native confidence on AUROC and CUR ($p < 0.05$), beat the per-image control on AUROC by at least 0.02 at $p < 0.05$, on DTU and on one dataset the head never saw. Table 5: pooled over 3, 5 and 10 views the head reaches AUROC 0.993 (VGGT) and 0.990 (Pi3X) against 0.772 and 0.820 for native confidence and 0.878 and 0.863 for the control; head minus control $+0.115$ and $+0.127$, head minus native $+0.221$ and $+0.171$, $p = 0.00005$ throughout and at each view count (Pi3 and MapAnything native: 0.841, 0.848). Native confidence lets half of the single-view pixels through a threshold that keeps 90% of supported ones; the head lets 2% through. The linear probe scores 0.989 and 0.985, so the information sits in the cross-view features. Reprojection consistency is the best training-free detector here (0.954, 0.914): real single-view geometry, unlike the fill, is cross-view incoherent; the head beats it by 0.04 and 0.08. The DTU conditions hold with five times the pre-registered margin. On CO3Dv2, which the head never saw, it keeps AUROC 0.929 (VGGT) and 0.932 (Pi3X) against 0.755 and 0.697 for native confidence, while the per-image control falls *below chance*, 0.461 and 0.426: whatever single-image cue it learned on the lab rig does not exist on phone video. Head minus control $+0.468$ and $+0.505$, head minus native $+0.174$ and $+0.234$, $p = 0.00005$ for all four. Reprojection consistency scores 0.794 and 0.888; the head beats it by 0.135 ($p = 0.0002$) on VGGT and by 0.044 ($p = 0.026$) on Pi3X, where both let 22% of single-view pixels through. Gate 3 passes on both datasets; ETH3D is pending: P4-vis-eth3d .

8. Limitations

Static scenes only. The benchmark labels are a proxy for real occlusion: a real unobserved region is not surrounded by a filled patch, and the stress tests bound but do not remove that gap. CO3Dv2 ground truth is COLMAP depth with 1 to 3% noise, so AbsRel below about 0.03 reads as equal, and three backbones saw CO3Dv2 in training, which can flatter their imagination there, not the leak. The benchmark cannot certify a detector trained on its own fills, as the fill-trained head shows. The visibility head needs multi-view ground truth to train; a variant trained on the support count of the backbone's own prediction, with no ground truth, is pending: P5-visself. It transfers from DTU to CO3Dv2; ETH3D is pending: P4-vis-eth3d. All findings depend on the released checkpoints.

9. Conclusion

When a feed-forward 3D model has not seen a surface, it mostly does not say so: a quarter to four fifths of the geometry it invents passes a confidence filter tuned to keep 90% of what it observed, and the leak grows when the occluder looks natural. The answer is not to stop the models from imagining but to make them say when they do. A support head trained on ground-truth visibility separates single-view from supported geometry at AUROC 0.99 where native confidence scores 0.8 and a one-image control 0.87; the same head trained on the benchmark's fills learns the fill, the warning that comes with masking labels.

LLM Usage Statement

A general-purpose LLM assistant was used, under the authors' direction, for parts of the code (the benchmark runner, the model wrappers, the run scripts and figure scripts), for literature lookup, and for editing the text of this paper, including grammar. The study design, the definitions and metrics, the choice and freezing of every hyperparameter, the execution and checking of every experiment, the verification of every reported number against its logged run, and the conclusions are the authors' own, and the authors are responsible for all content. The assistant was not used to generate results, figures from unlogged data, or references; every citation was checked against the published record. The tool is named in the camera-ready version.

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